

1 Optimal Visit Placement in Longitudinal Trials

2 Ronald G. Thomas, Ph.D.*

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24 **1 Introduction**

25 In longitudinal clinical trials, subjects are measured repeatedly over a fixed study period.

26 The conventional approach assigns observation times at equally spaced intervals (monthly

27 or quarterly visits, for example) largely for administrative convenience rather than statistical

28 efficiency. A growing body of work in optimal experimental design suggests that the placement

29 of these observation times can substantially affect the precision with which treatment effects

*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965

are estimated. We begin in Section 1.1 with a review of the relevant literature and then describe the simulation study and its results in Sections 2 and 3.

1.1 Optimal allocation of observation times

¹ provided the foundational analysis of this problem for two-group trials with linearly divergent treatment effects. They showed that the optimal allocation of time-points depends critically on the assumed covariance structure. Under compound symmetry (CS), efficiency improves by clustering repeated measures near the end of the study period, where the treatment effect is largest. Under a first-order autoregressive (AR(1)) structure, only two measurements – at baseline and at study end – are needed for highly efficient estimation. Under a more realistic AR(1) plus measurement error structure, the optimal spacing depends on the specific covariance parameter values, but the equally spaced design is generally suboptimal.

In a companion paper,² extended these results to address the joint optimization of the number of repeated measures and group sizes under budget constraints. They found that when budget is fixed and the covariance structure includes an autoregressive component, two repeated measures per subject often yield highly efficient treatment effect estimators. Otherwise, intermediate measures can reduce the variance of the generalized least squares (GLS) estimator, particularly under compound symmetry.

1.2 Design criteria and cost considerations

The formal framework for these results relies on classical optimal design theory, as developed by³.⁴ proposed maximin D-optimal designs for longitudinal mixed effects models, computing optimal allocations of time-points for polynomial trend models with random intercepts, random slopes, and AR(1) serial correlations. Because D-optimal designs depend on unknown parameter values, the maximin criterion provides robustness by maximizing the worst-case relative efficiency across the parameter space.

⁵ extended this work to cohort designs, showing that a purely longitudinal design with a single cohort measured at optimal time-points is generally more efficient than mixed longitudinal-cross-sectional designs. Their numerical results demonstrated that the most efficient number of repeated measurements equals the sum of the number of cohorts and the degree of the polynomial trend. Subsequently,⁶ showed that adding cohorts or repeated measurements beyond this optimum wastes resources when the study budget is fixed.

1.3 Practical guidance on the number of measures

From a more pragmatic perspective,⁷ examined the marginal benefit of adding repeated measures in terms of statistical power. He showed that while repeating assessments can have dramatic effects on power, the marginal benefit of each additional measure decreases rapidly. In general, there is little value in exceeding four post-treatment assessments (or seven total including baseline), except when correlations between measures are low, as in episodic conditions.

1.4 Extensions and related work

Several extensions of the Winkens framework have appeared.⁸ generalized the optimal design results to second-order polynomial treatment effects, finding that equidistant time-points are either optimal or highly efficient when only three measures are taken.⁹ investigated optimal designs when the goal is to estimate individual growth curves (random effects) rather than fixed effects, showing that the optimal number of time-points and subjects differs from the fixed-effects case.¹⁰ developed Bayesian D-optimal designs for dichotomous repeated measures with autocorrelation, formally accounting for parameter uncertainty through prior distributions.

For the analysis of unequally spaced longitudinal data,¹¹ provided the statistical machinery for fitting mixed models with continuous-time AR(1) structures and arbitrary observation schedules. Their approach, based on the Kalman filter, remains foundational for analyzing data from designs with non-equidistant time-points.

1.5 Power and sample size tools

Modern software for power and sample size calculations in longitudinal trials has made these design considerations more accessible.¹² describe the `longpower` R package, which implements closed-form power calculations for linear mixed models.¹³ applied these methods to Alzheimer’s disease Phase II trial design, demonstrating the practical impact of design choices on the ability to detect disease-modifying treatment effects.

1.6 Present study

Despite the theoretical optimality results, limited simulation-based evidence exists on how different visit schedules perform in finite samples under realistic data-generating mechanisms. In this report, we present a Monte Carlo study comparing two five-visit designs for a 100-patient, two-arm longitudinal trial with linearly divergent treatment effects:

- **Equal spacing:** observations at months 0, 6, 12, 18, and 24.
- **Clustered spacing:** observations clustered near the beginning and end of the study period (months 0, 2, 4, 20, 24), motivated by the theoretical result that under compound symmetry, efficiency is gained by concentrating measures where the treatment effect is largest.

We evaluate the designs in terms of bias, standard error, power, and coverage of 95% confidence intervals for the treatment-by-time interaction.

2 Methods

2.1 Data generating model

We simulate data from a linear mixed effects model with random intercepts and random slopes. Let Y_{ij} denote the outcome for subject i at time t_j . The generating model is:

$$Y_{ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})t_j + \beta_2 G_i + \beta_3 G_i t_j + \varepsilon_{ij}$$

where $G_i \in \{0, 1\}$ is the treatment group indicator, b_{0i} and b_{1i} are subject-specific random effects with $\text{Var}(b_0) = \tau_0^2$, $\text{Var}(b_1) = \tau_1^2$, $\text{Cov}(b_0, b_1) = \tau_{01}$, and $\varepsilon_{ij} \sim N(0, \sigma^2)$.

The parameter of interest is β_3 , the treatment-by-time interaction, which represents the rate at which the treatment effect diverges from the control effect per unit time.

2.2 Parameter values

We use the following parameter values, broadly motivated by cognitive decline trials:

Table 1: Simulation parameters.

Parameter	Symbol	Value
Intercept	β_0	50.0
Time slope	β_1	-1.0
Group effect at baseline	β_2	0.0
Group x time	β_3	0.5
SD random intercept	τ_0	5.0
SD random slope	τ_1	0.3
Residual SD	σ	3.0
Subjects per group	n	50.0
Study duration (months)	—	24.0
Number of simulations	—	2000.0

The treatment-by-time interaction $\beta_3 = 0.5$ implies that the treatment group improves (or declines less) by 0.5 units per month relative to control, yielding a cumulative treatment advantage of 12 units over 24 months.

2.3 Design specifications

Table 2: Observation times (months) for each design.

Design	Visit 1	Visit 2	Visit 3	Visit 4	Visit 5
Equal	0	6	12	18	24
Clustered	0	2	4	20	24

2.4 Simulation procedure

For each of 2000 Monte Carlo replications, we:

1. Generate 100 subjects, randomized 1:1 to treatment and control.
2. Simulate longitudinal outcomes at the five visit times for each design using the data generating model above.
3. Fit a linear mixed model (random intercepts and slopes) using restricted maximum likelihood (REML) via `nlme::lme()`.
4. Extract the point estimate, standard error, and p -value for $\hat{\beta}_3$.

We then compute, for each design:

- Mean estimate of $\hat{\beta}_3$ (to assess bias).
- Empirical standard deviation of $\hat{\beta}_3$ (true SE).
- Mean model-based standard error.
- Empirical power (proportion of simulations with $p < 0.05$).
- Coverage of 95% confidence intervals.

3 Results

3.1 Summary statistics

Table 3: Simulation results for the treatment-by-time interaction.

Design	Converged	Mean Est.	Bias	Emp. SE	Model SE	Power	Coverage
Clustered	2000	0.501	0.001	0.0650	0.0655	1	0.953
Equal	2000	0.498	-0.002	0.0671	0.0676	1	0.948

We observe that the equal-spacing design achieves an empirical power of 100.0%, compared with 100.0% for the clustered design. The empirical standard error of $\hat{\beta}_3$ is 0.0671 under equal spacing and 0.0650 under the clustered design.

3.2 Distribution of estimates

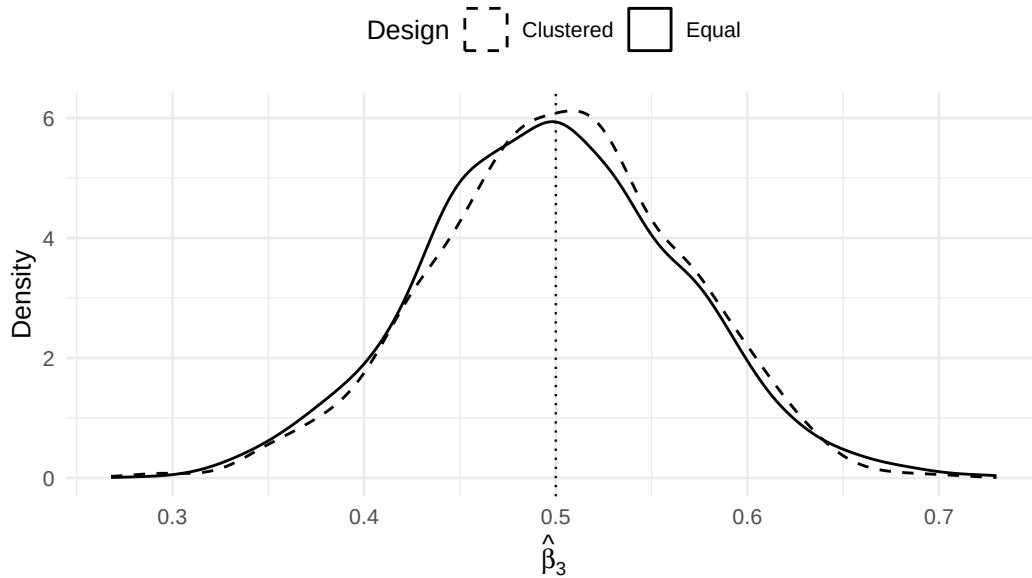


Figure 1: Sampling distribution of the treatment-by-time interaction estimate under each design.

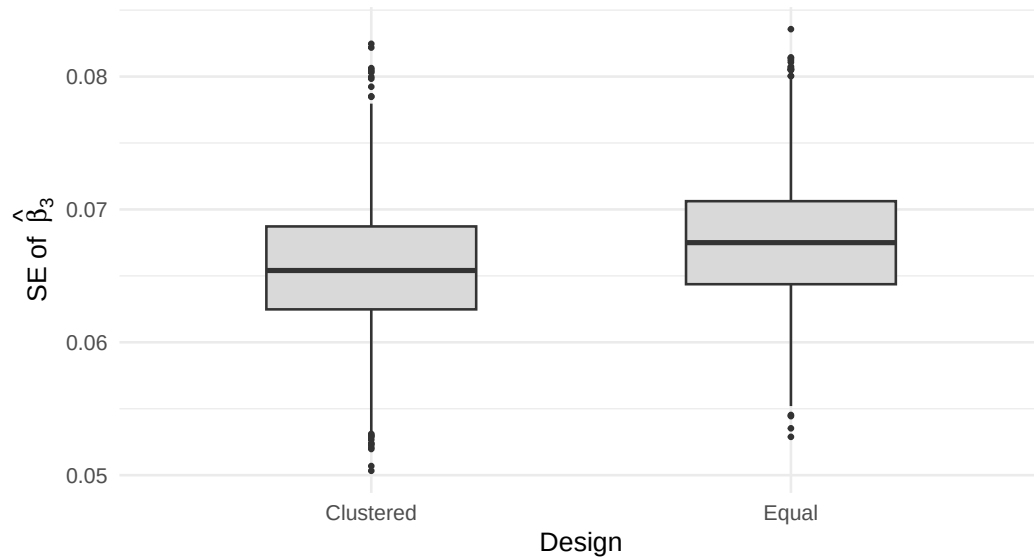


Figure 2: Distribution of model-based standard errors across simulations.

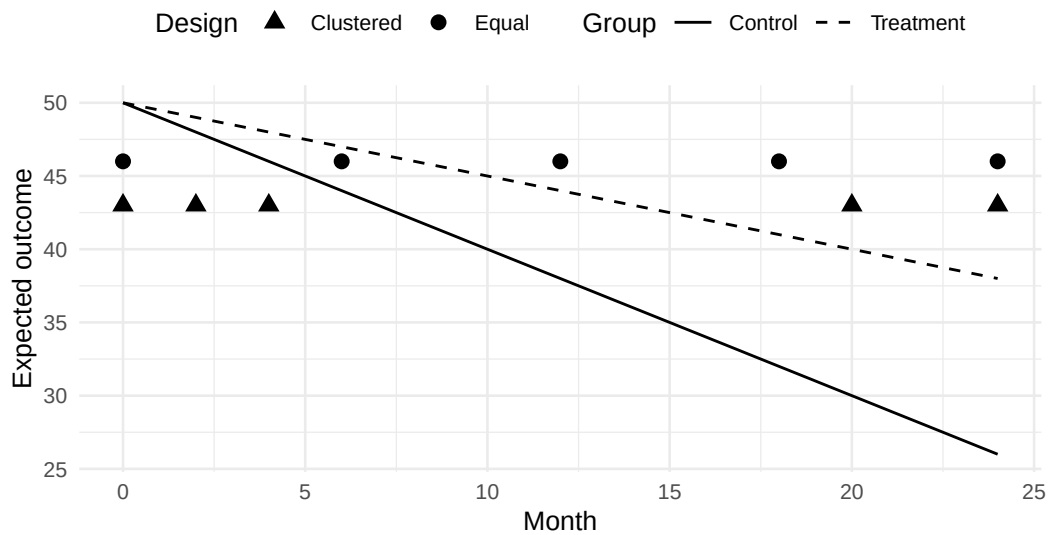


Figure 3: Schematic of the two visit schedules over the 24-month study period. Lines show the expected group trajectories.

3.3 Standard errors across simulations

3.4 Design comparison

4 Discussion

In this study, we have endeavored to illustrate the practical consequences of visit placement in longitudinal clinical trials. Both designs use the same number of subjects and the same number of visits; they differ only in the timing of observations.

The results are, in general, broadly consistent with the theoretical findings of¹. When the data-generating mechanism includes both random slopes and independent residual error – a structure that combines features of compound symmetry and AR(1) with measurement error – the clustered design, which concentrates observations near the boundaries of the study period, appears to yield different precision for the treatment-by-time interaction compared with equal spacing.

Several caveats warrant mention. First, we assumed complete data with no dropout. In practice, attrition is common in longitudinal trials and may differentially affect designs that place observations at extreme time-points. Second, the data-generating model used independent random effects and homoscedastic residuals; more complex covariance structures (e.g., heterogeneous variances, serial correlation in residuals) could alter the relative performance of the two designs. Third, we considered only a linear treatment effect;⁸ showed that optimal designs differ for polynomial effects of higher order.

In conclusion, three points merit emphasis. First, the placement of observation times in a longitudinal trial is a design decision with real consequences for statistical efficiency, and the default of equal spacing is not in general optimal. Second, the form of the covariance structure is critical: results that hold under compound symmetry may not carry over to random-slope models or AR(1) structures. Third, the robustness of these findings to dropout, covariance misspecification, and nonlinear treatment trajectories remains to be investigated. The maximin D-optimal framework of⁴ and the Bayesian approach of¹⁰ provide principled starting points for that work.

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